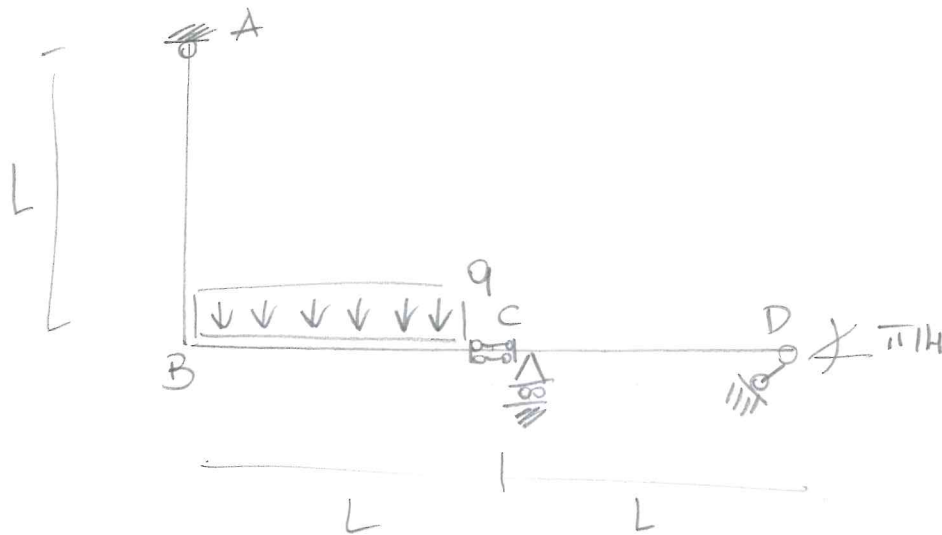


CARATTERISTICHE della SOLLECITAZIONE



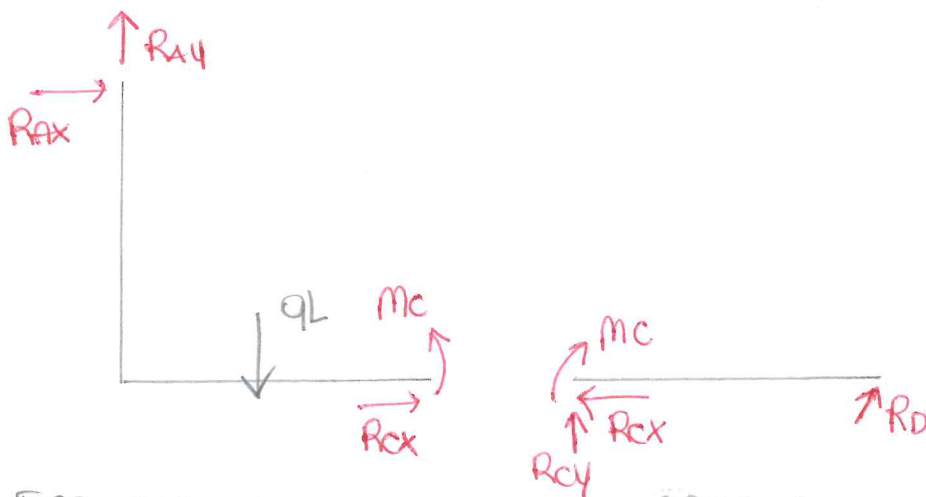
$$t=2 \quad s=\sum MS=2+2+1+1$$

$$3t-s=l-1$$

$$6-6=0 \quad l=1$$

$$\sum C \Rightarrow l=0 \rightarrow \text{ISOSTATICA}$$

$$\underline{e_D} = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right)$$



EC5 CORPO 1

$$ex \parallel R_{Ax} + R_{Cx} = 0$$

$$ey \parallel R_{Ay} - qL = 0$$

$$ez \parallel_A = MC - qL \cdot \frac{L}{2} + R_{Cx} \cdot L = 0$$

CORPO 2

$$ex \parallel -R_{Cx} + R_D \frac{\sqrt{2}}{2} = 0$$

$$ey \parallel R_{Cy} + R_D \frac{\sqrt{2}}{2} = 0$$

$$ez \parallel_D = -MC - R_{Cy} \cdot L = 0$$

$$R_{Ay} = qL \quad R_{Ax} = -R_{Cx}$$

$$\begin{cases} MC = qL \frac{L}{2} - R_{Cx} \cdot L \\ MC = -R_{Cy} \cdot L \end{cases}$$

$$R_{Cx} = +R_D \frac{\sqrt{2}}{2} \rightarrow R_{Cx} = -R_{Cy}$$

$$R_{Cy} = -R_D \frac{\sqrt{2}}{2}$$

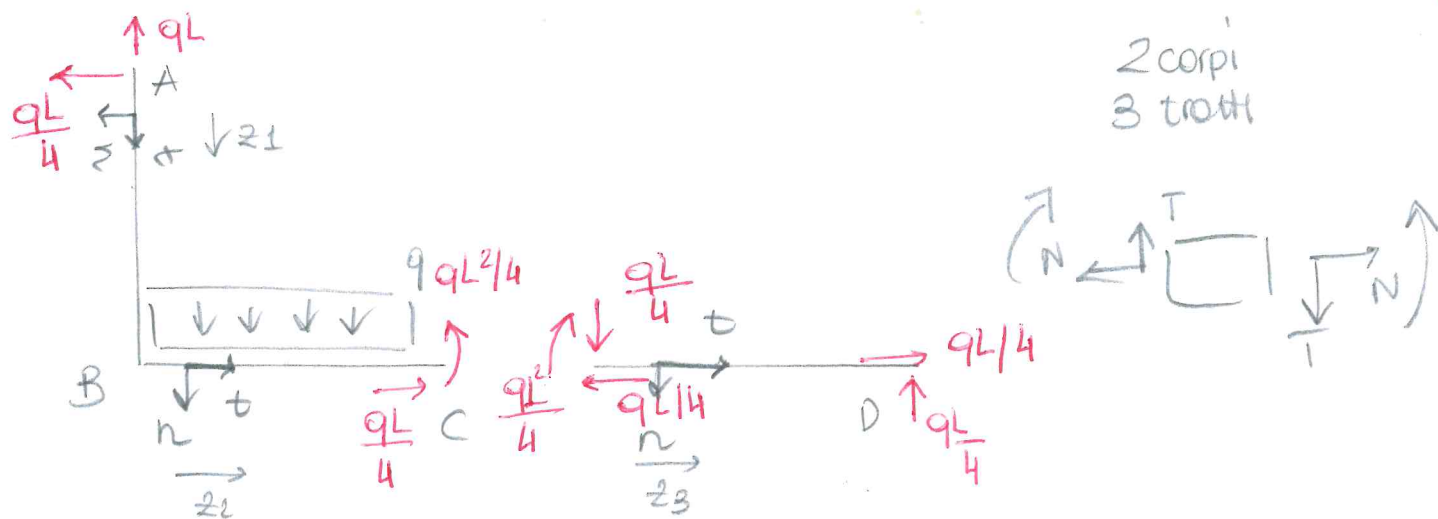
$$0 = \frac{qL^2}{2} - R_{Cx}L + \underbrace{R_{Cy}L}_{R_{Cx} = -R_{Cy}}$$

$$\frac{qL^2}{2} = 2R_{Cx}L \quad R_{Cx} = \frac{qL}{4}$$

$$R_{Ax} = -\frac{qL}{4} \quad R_{Ay} = qL \quad MC = \frac{qL^2}{4} \quad R_{Cy} = -\frac{qL}{4} \quad R_{Cx} = \frac{qL}{4}$$

$$R_D = \frac{qL}{4} \cdot \frac{2}{\sqrt{2}} \quad \frac{qL}{4} \cdot \frac{\sqrt{2}}{2\sqrt{2}} = \frac{qL\sqrt{2}}{4}$$

$$\left\{ \begin{array}{l} R_{Dx} = \frac{qL\sqrt{2}}{4} \cdot \frac{\sqrt{2}}{2} = \frac{qL}{4} \\ R_{Dy} = qL/4 \end{array} \right.$$



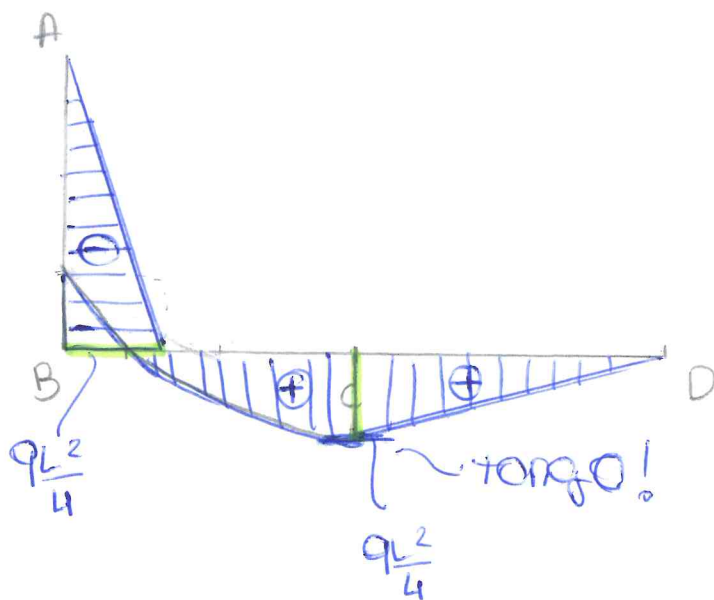
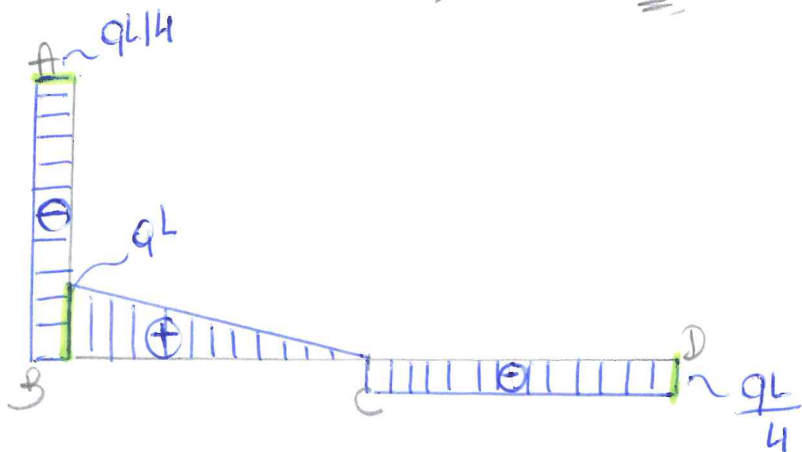
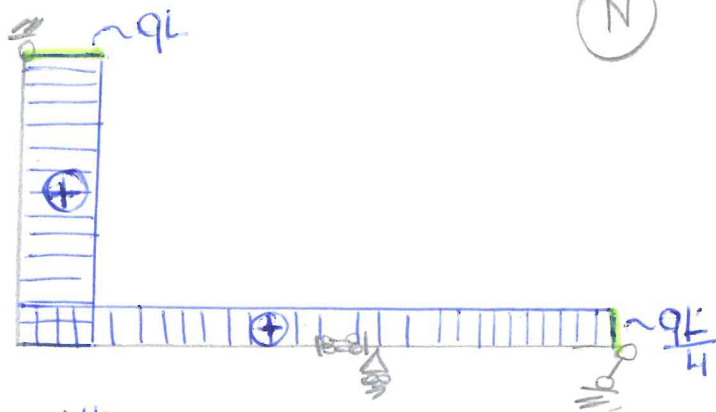
$$z_1 \in (0, L) \quad z_2 \in (0, L) \quad z_3 \in (0, L)$$

$$N(z) = \begin{cases} qL & \text{su AB } z_1 \in (0, L) \text{ pr.} \\ qL/4 & \text{su BC } z_2 \in (0, L) \text{ seg.} \\ qL/4 & \text{su CD } z_3 \in (0, L) \text{ seg.} \end{cases}$$

$$T(z) = \begin{cases} -\frac{qL}{4} & \text{su AB } z_1 \in (0, L) \text{ pr.} \\ qL - qz & \text{su BC } z_2 \in (0, L) \text{ a prec. / a seguire } q(L-z) \\ -qL/4 & \text{su CD } z_3 \in (0, L) \text{ seg.} \end{cases}$$

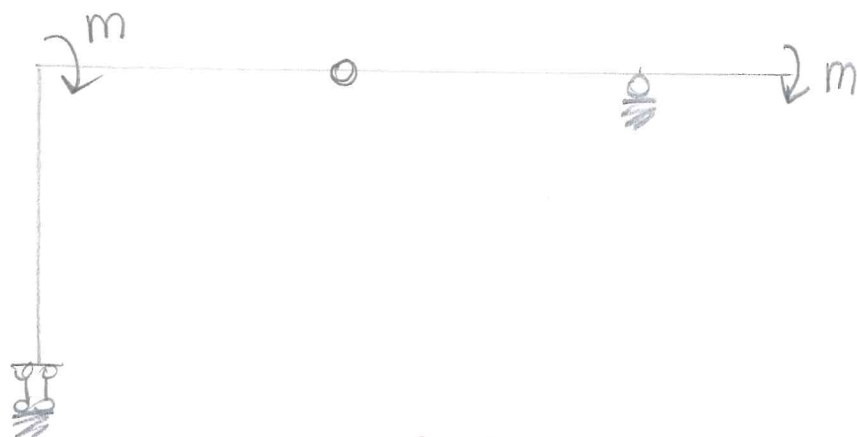
$$M(z) = \begin{cases} -\frac{qL}{4} \cdot z & \text{su AB } z_1 \in (0, L) \text{ pr.} \\ qL^2/4 - q(L-z)^2 & \text{su BC } z_2 \in (0, L) \text{ seg.} \\ \frac{qL}{4}(z-L) & \text{su CD } z_3 \in (0, L) \text{ seg.} \end{cases}$$

(N)



$$M_{BC} = 0 \text{ in } z = L\left(1 - \frac{1}{\sqrt{2}}\right)$$

ESEMPIO 5

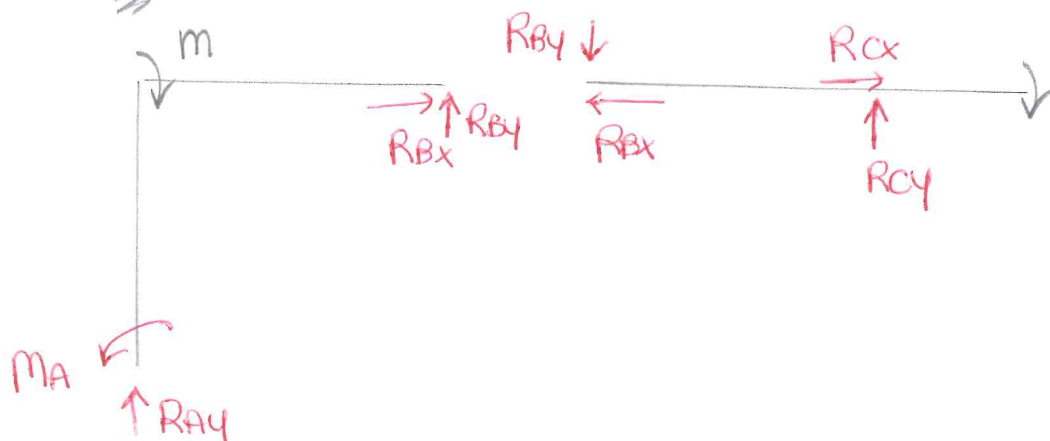


$$t = 2$$

$$S = 2 + 2 + 2 = 6$$

$$3t - S = 2 - 1$$

$$2 = 1 \rightarrow AC \quad \begin{matrix} 2 = 0 \\ 1 = 0 \end{matrix}$$



ECS CORPO 1

$$e_{x1} \quad R_{Bx} = 0$$

$$e_{y1} \quad R_{Ay} + R_{By} = 0$$

$$e_{\theta A} \quad M_A - m + R_{By} \cdot L$$

CORPO 2

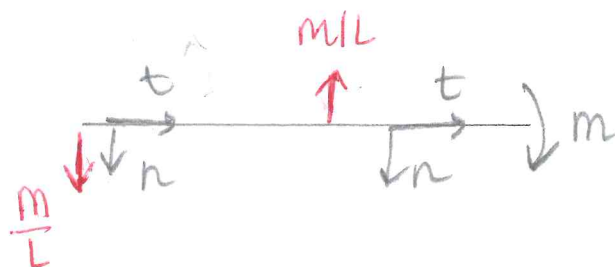
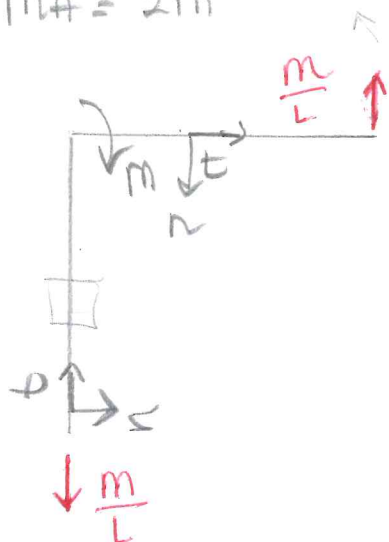
$$-R_{Bx} + R_{Cx} = 0$$

$$-R_{By} + R_{Cy} = 0$$

$$e_{\theta B} \quad R_{Cy} \cdot L - m = 0$$

$$R_{Bx} = 0 \quad R_{Cx} = 0 \quad R_{Cy} = +\frac{m}{L} \quad R_{By} = +\frac{m}{L} \quad R_{Ay} = -\frac{m}{L}$$

$$M_A = 2m$$



CDS 2 corpi e tratti matematici (Equilibrio rispetto
alla terza
locomotiva)

$$AB \quad z_1(0, L)$$

$$N_{AB}(z_1) = +\frac{m}{L}$$

$$T_{AB}(z_1) = 0$$

$$M_{AB}(z_1) = 0$$

$$BC \quad z_2(0, L)$$

$$N_{BC}(z_2) = 0$$

$$T_{BC}(z_2) = -\frac{m}{L}$$

$$M_{BC}(z_2) = \frac{m}{L}(L - z_2)$$

$$CD \quad z_3(0, L)$$

$$N_{CD}(z_3) = 0$$

$$T_{CD}(z_3) = -\frac{m}{L}$$

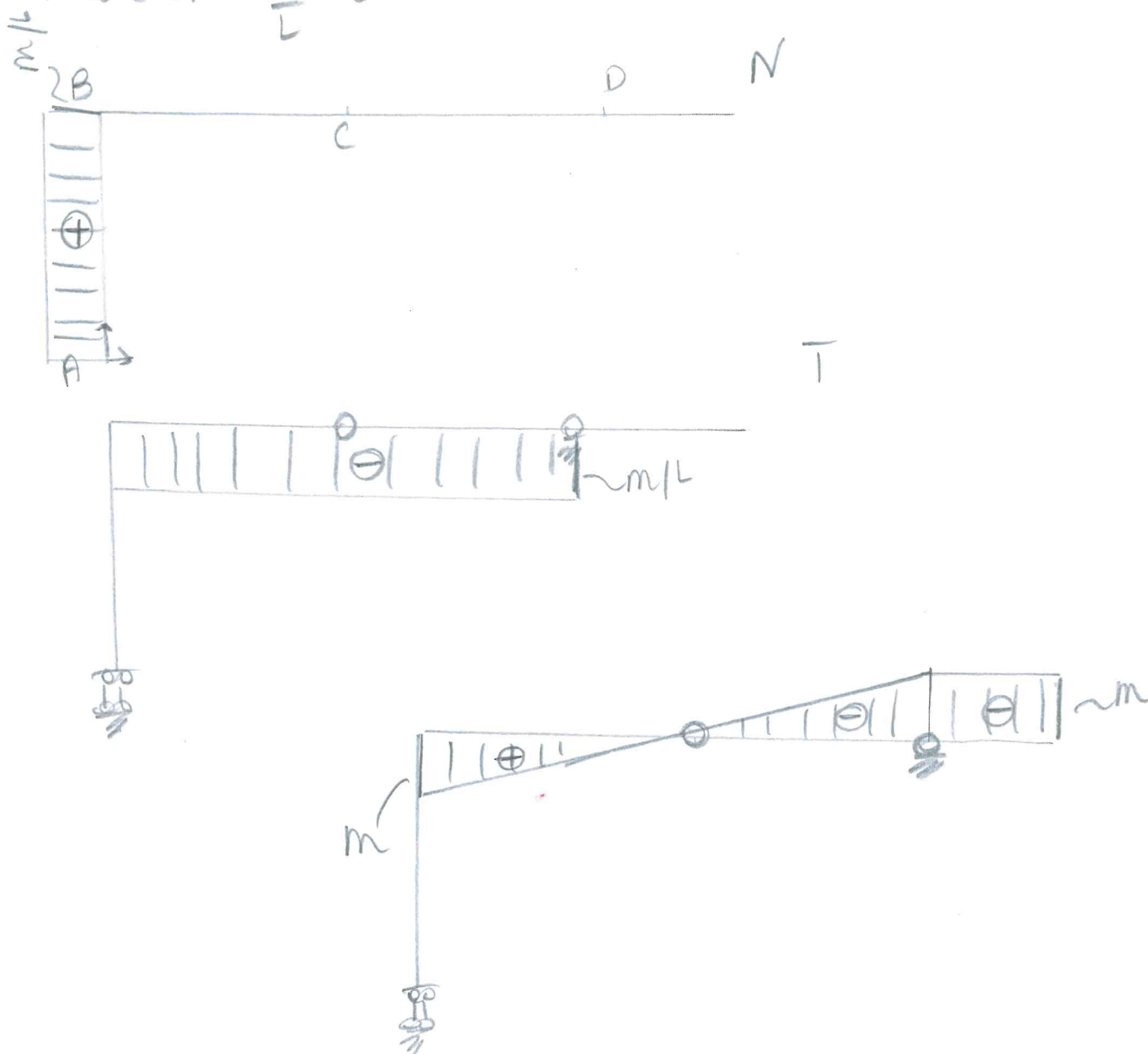
$$M_{CD}(z_3) = -\frac{m}{L}z_3$$

$$DE \quad z_4(0, L/2)$$

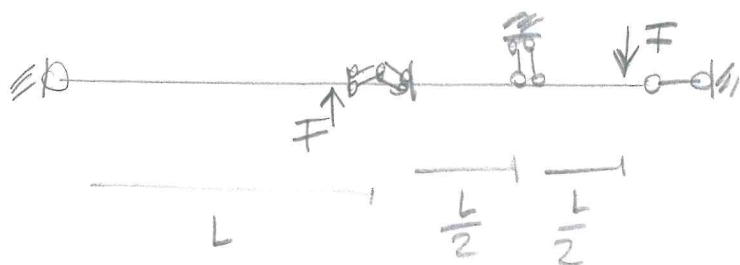
$$N_{DE}(z_4) = 0$$

$$T_{DE}(z_4) = 0$$

$$M_{DE}(z_4) = -m$$



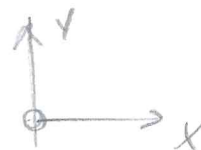
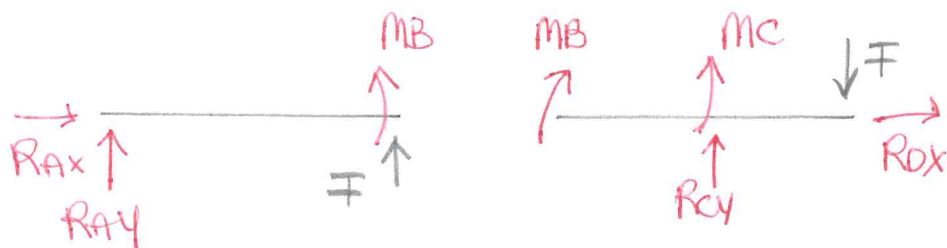
ESEMPLO 3



$$t=2 \quad S=2+1+2+1$$

$$3t-S = 2-1$$

$$l=1 \quad \neq C \quad l=0 \quad 1=0$$



ECS

1 CORPO

$$ex|| \quad R_{AX} = 0$$

$$ey|| \quad R_{AY} + F = 0$$

$$ez||_A \quad M_B + F \cdot L = 0$$

2 CORPO

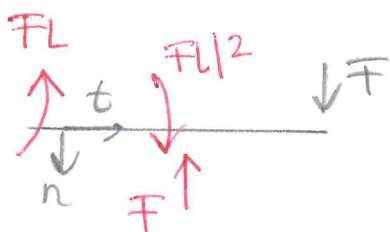
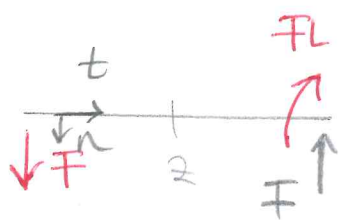
$$ex|| \quad R_{DX} = 0$$

$$ey|| \quad R_{CY} - F = 0$$

$$ez||_D \quad -M_B + M_C - R_{CY} \cdot \frac{L}{2} = 0$$

$$R_{AX} = 0 \quad R_{DX} = 0 \quad R_{AY} = -F \quad M_B = -FL \quad R_{CY} = F$$

$$M_C = M_B + R_{CY} \cdot \frac{L}{2} = -FL + F \cdot \frac{L}{2} = -\frac{FL}{2}$$



2 corpi 3 tratti matematici



$$AB \quad z_1 \in (0, L)$$

$$N_{AB}(z_1) = 0$$

$$T_{AB}(z_1) = -F$$

$$M_{AB}(z_1) = -Fz$$

$$BC \quad z_2 \in (0, L/2)$$

$$N_{BC}(z_2) = 0$$

$$T_{BC}(z_2) = 0$$

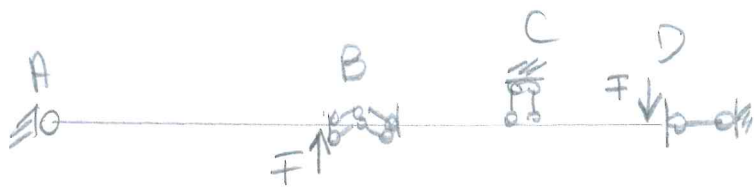
$$M_{BC}(z_2) = -FL$$

$$CD \quad z_3 \in (0, L/2)$$

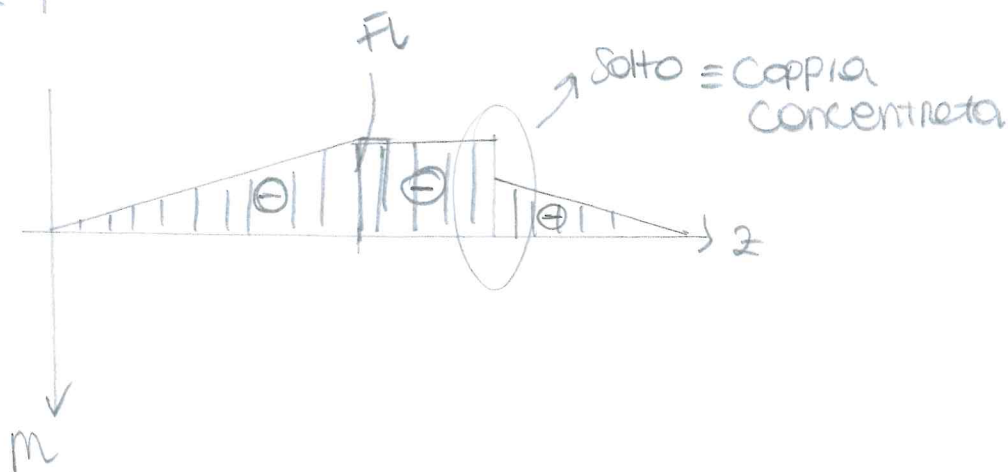
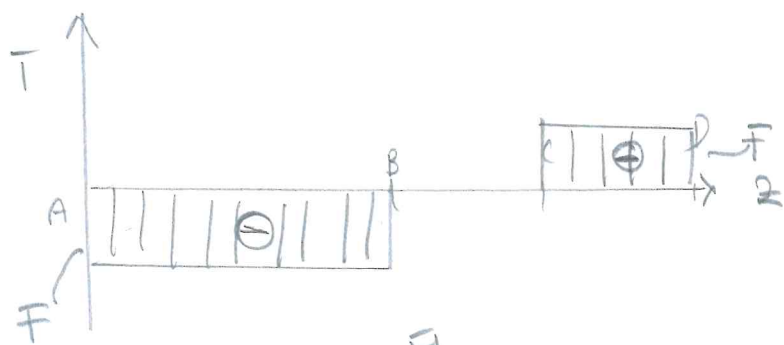
$$N_{CD}(z_3) = 0$$

$$T_{CD}(z_3) = +F$$

$$M_{CD}(z_3) = -F(\frac{L}{2} - z_3)$$



$$N = 0 \quad \forall z \in (0, 2L)$$



CARATTERISTICHE della SOLLECITAZIONE:

$t=2 \quad S=\sum MS=2+2+1$
 $\quad \quad \quad =5$

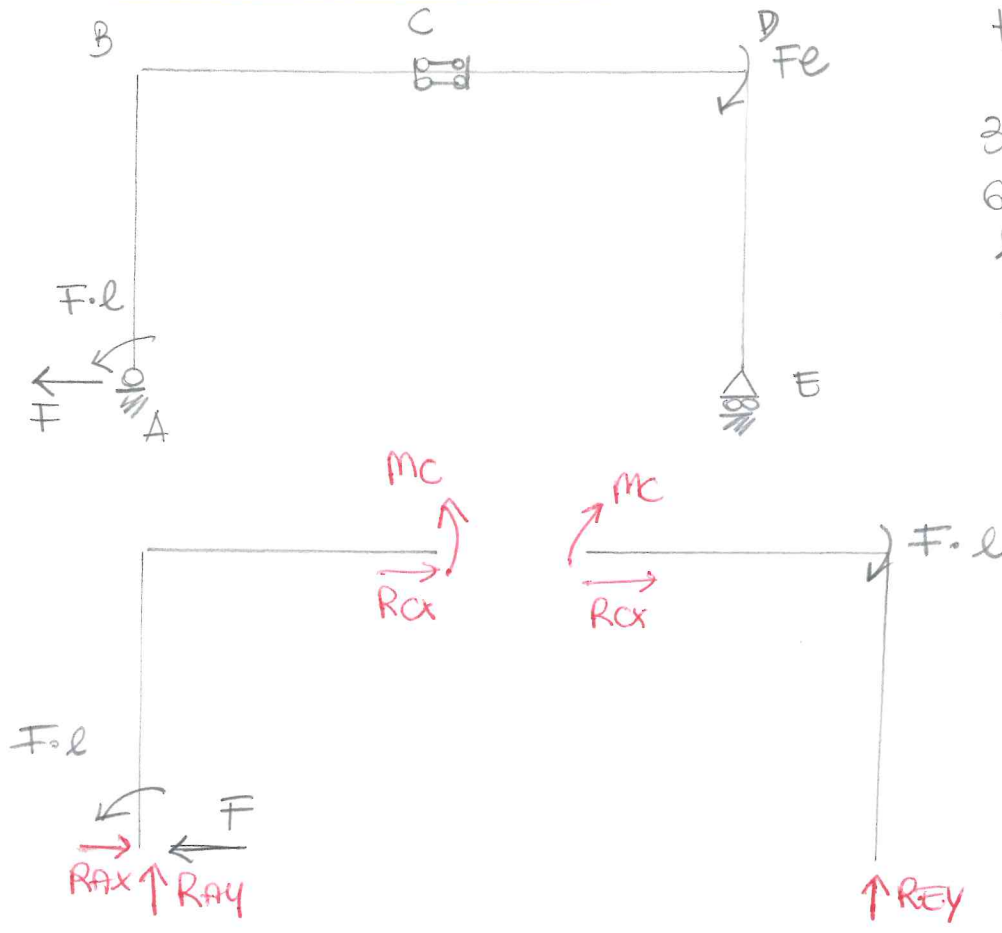
$3t-S=l-i$

$6-5=1$

$l-i=1$

$\exists C \rightarrow l=1 \quad i=0$

str. labile non iperstatica



EOS CORPO 1

$ex|| \quad R_{AX} - F + R_{CX} = 0$

$ey|| \quad R_{AY} = 0$

$ez||_A \quad FL + MC - R_{CX}L = 0$

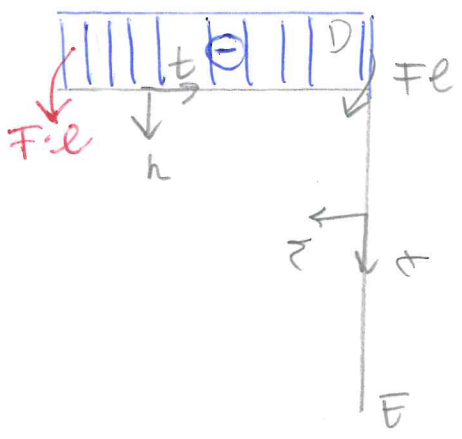
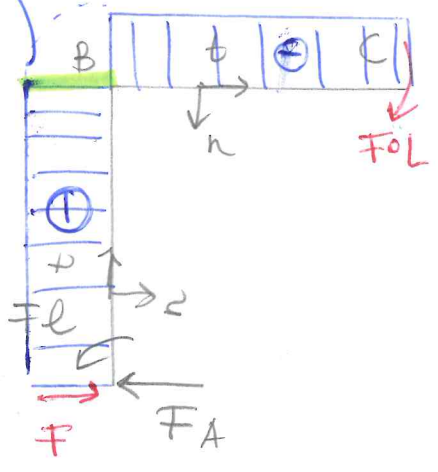
CORPO 2

$ex|| \quad -R_{CX} = 0$

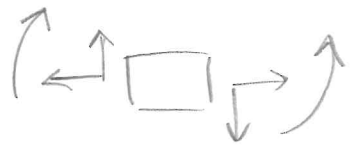
$ey|| \quad R_{EY} = 0$

$ez||_C \quad R_{EY} \cdot L - MC + FL = 0$

$FE \quad R_{AX} = F \quad R_{AY} = 0 \quad R_{CX} = 0 \quad MC = -FL \quad RE = 0$



- 21 ∈ (AB)
- 22 ∈ (BC)
- 23 ∈ (CD)
- 24 ∈ (DE)



$$N_{AB}(z_1) = 0 \quad z_1 \in (0, L)$$

$$N_{BC}(z_2) = 0 \quad z_2 \in (0, L) \quad \rightarrow \quad N(z) \stackrel{\downarrow}{=} 0$$

$$N_{CD}(z_3) = 0 \quad z_3 \in (0, L) \quad T(z) = 0$$

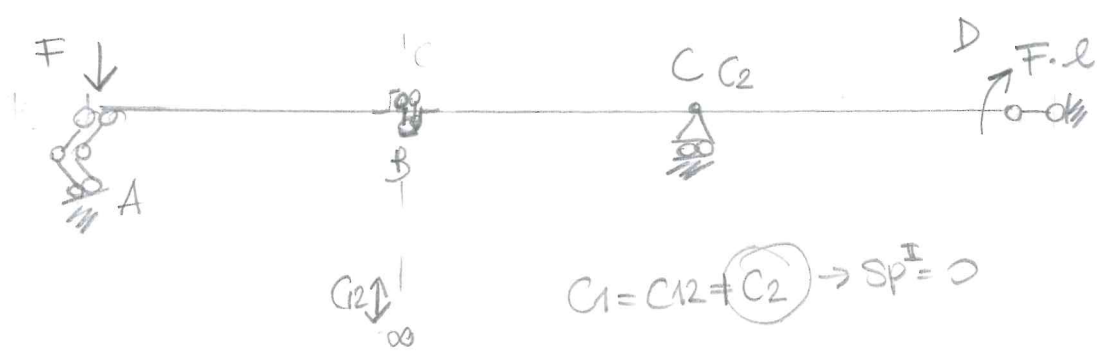
$$N_{DE}(z_4) = 0 \quad z_4 \in (0, L)$$

$$M_{AB}(z) = -Fl \quad z_1 \in (0, L)$$

$$M_{BC}(z) = -Fl \quad z_2 \in (0, L)$$

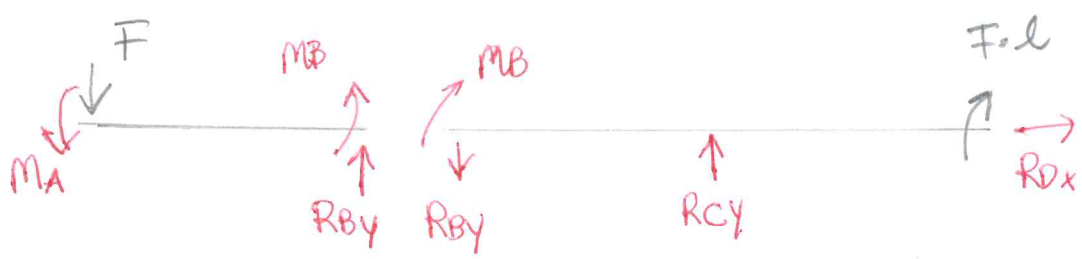
$$M_{CD}(z) = -Fl \quad z_3 \in (0, L)$$

$$M_{DE}(z) \stackrel{\downarrow}{=} 0$$



$t=2$
 $s = \sum MS = 1+1+1+2 = 5$
 $3t-s = 6-5 = 1$
 $\exists C \quad l=1 \quad 1=0$
 $TSV = 0 \text{ ok}$

$C_1 = C_{12} = C_2 \rightarrow SP^I = 0$



ECS (1)

ex|| 0
ey|| $-F + RBy = 0$
ez||_A $RBy \cdot L + MA + MB = 0$

ECS (2)

ex|| $Rdx = 0$
ey|| $-RBy + Rcy = 0$
ez||_B $-MB + Rcy \cdot L - F \cdot l = 0$

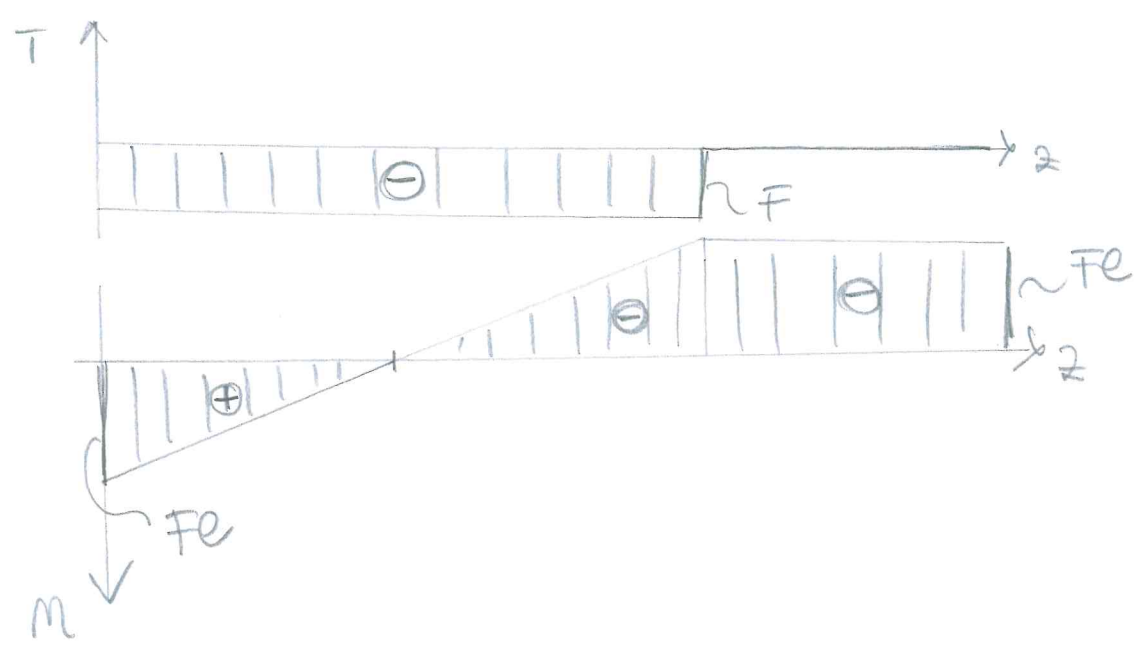
$RBy = F \quad Rcy = F \quad MB = Fl - Fe = 0$
 $MA = -Fl$



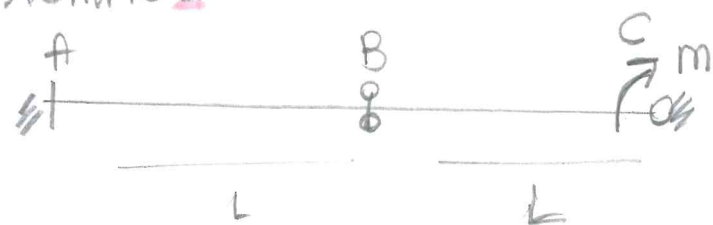
$N(z) = 0$ su AB, BC, CD

$T(z) = \begin{cases} -F & \text{su AB} \\ -F & \text{su BC} \\ 0 & \text{su CD} \end{cases}$

$M(z) = \begin{cases} Fe - Fl & \text{su AB } (0, L) \\ -Fl + F(L-z) & \text{su BC } (0, L) \\ -Fl & \text{su CD } (0, L) \end{cases}$
 $z=L \quad -Fl$

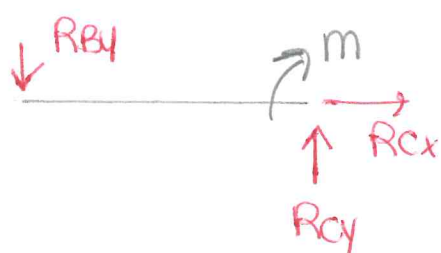
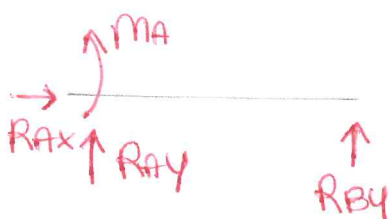


ESEMPIO 2



$$t=2 \quad S=6 \quad 3t-S=l-1$$

$$l=i \quad \cancel{AC} \quad \begin{matrix} l=0 \\ 1=0 \end{matrix}$$



ECS

1 CORPO

ex|| $R_{Ax} = 0$

ey|| $R_{Ay} + R_{By} = 0$

ez||_A $-R_{By} \cdot L + M_A = 0$

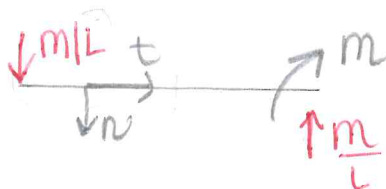
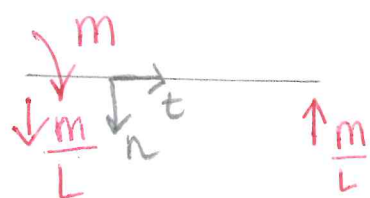
2 CORPO

$R_{Cx} = 0$

$-R_{By} + R_{Cy} = 0$

et||_C $R_{By} \cdot L - m = 0$

$$R_{Ax} = 0 \quad R_{Cx} = 0 \quad R_{By} = \frac{m}{L} \quad R_{Cy} = \frac{m}{L} \quad M_A = m \quad R_{Ay} = -\frac{m}{L}$$



metodo diretto concio

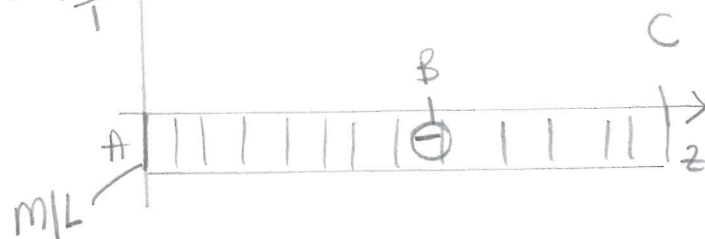
2 corpi = 2 tratti mot.

$$N_{AB}(z_1) = 0$$

$$T_{AB}(z_1) = -\frac{m}{L}$$

$$M_{AB}(z_1) = m - \frac{m}{L} \cdot z_1$$

$$AB \rightarrow z_1 \in (0, L) \quad \uparrow T$$

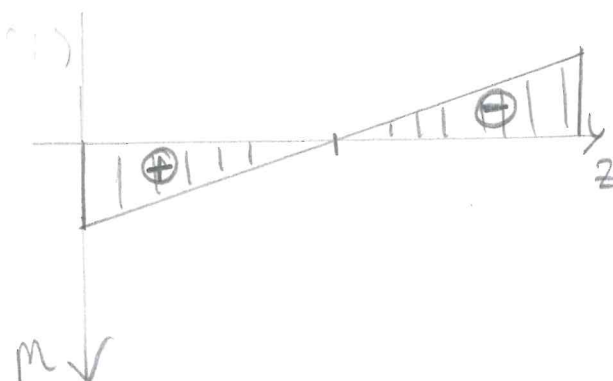


$$N_{BC}(z_2) = 0$$

$$T_{BC}(z_2) = -\frac{m}{L}$$

$$M_{BC}(z_2) = -\frac{m}{L} z_2$$

$$BC \rightarrow z_2 \in (0, L)$$



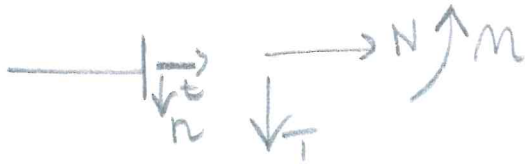
Eq. rispetto (t, n, ex)

$$1) N_{AB}(z_1) = 0$$

$$\frac{m}{L} + T_{AB}(z_1) = 0$$

$$-m + \frac{m}{L} \cdot z + M_{AB}(z) = 0$$

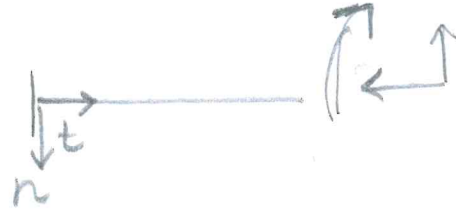
← a proceotere



$$2) N_{BC}(z_2) = 0$$

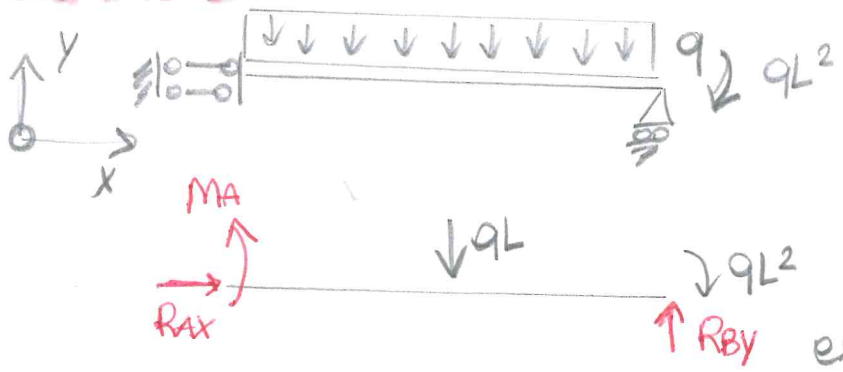
$$T_{BC}(z_2) + \frac{m}{L} = 0$$

$$m - \frac{m}{L} z + M_{BC}(z_2) = 0$$



CARATTERISTICHE DELLA SOLLECITAZIONE

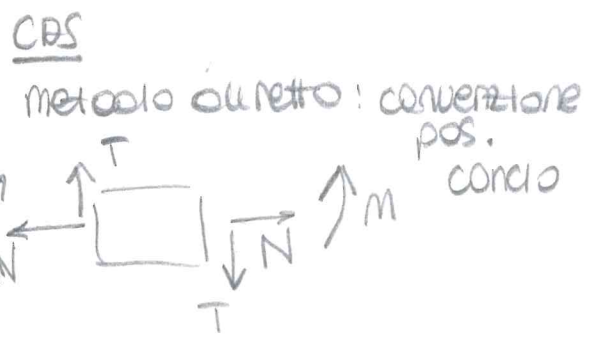
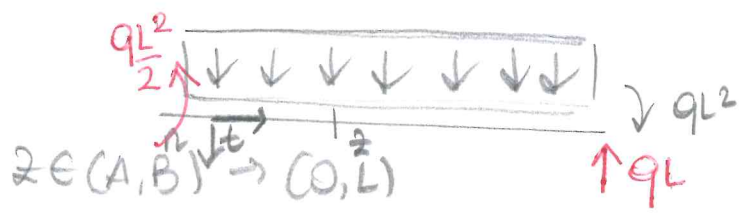
ESEMPIO 1



$t=1 \quad s=3 \quad 3t-s=2-1$
 $\sum C \quad l=0$
 $\quad \quad \quad 1=0$

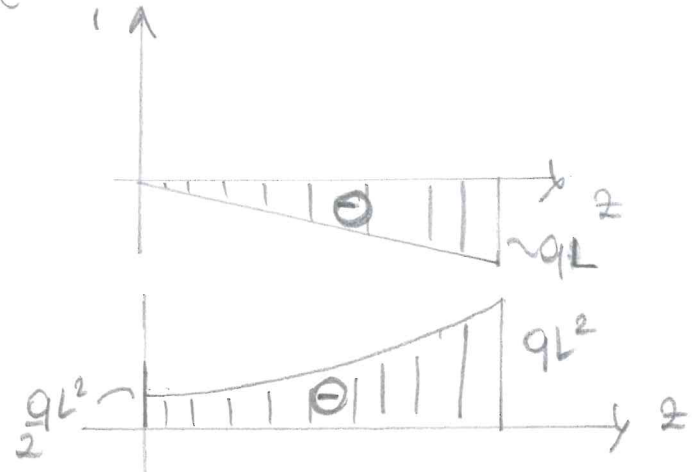
$R_{ax} = 0$
 $R_{by} = qL$
 $M_A = q \frac{L^2}{2} - qL^2 + qL^2 = 0$

Eq. in C.S.
ex II $R_{ax} = 0$
eq II $-qL + R_{by} = 0$
eq III $M_A - qL \cdot \frac{L}{2} + R_{by} \cdot L - qL^2 = 0$



$N_{AB}(z) = 0$
 $T_{AB}(z) = -q \cdot z$
 $M_{AB}(z)|_z = -q \frac{L^2}{2} - qz \cdot \frac{z}{2} \rightarrow -\frac{q}{2}(L^2 + z^2)$

Equilibrio (a precoltura)
 $N_{AB} = 0$
 $q \cdot z + T_{AB}(z) = 0$
 $q \frac{L^2}{2} + qz \cdot z + M_{AB}(z) = 0$



Eq. in INDEFINITE di EQUILIBRIO! m ✓

$N'(z) + p(z) = 0$
 $T'(z) + q(z) = 0$
 $M'(z) - T(z) + m(z) = 0$

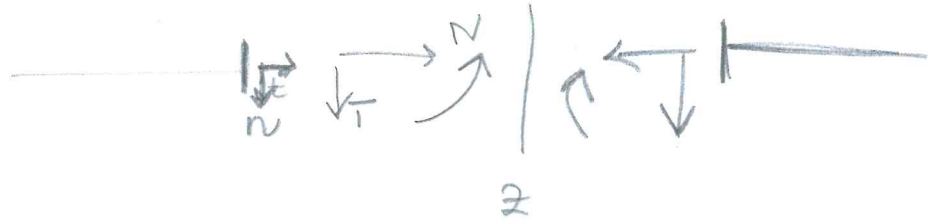
→ $N'(z) = 0$ integr. $N(z) = C_1$
 $T'(z) = -q$ → $T(z) = -qz + C_2$
 $M'(z) = T(z)$ → $M(z) = -q \frac{z^2}{2} + C_2 z + C_3$

+ BCs (STATIC)

$$T_{AB}(0) = 0$$

$$N_{AB}(L) = 0$$

$$M_{AB}(L) = -qL^2$$



$$C_1 = 0 \quad T_{AB}(x) = -qx + C_2 \rightarrow x=0 \quad C_2 = 0$$

$$M_{AB}(L) = -qL^2 = -q\frac{L^2}{2} + C_3 \quad C_3 = -q\frac{L^2}{2}$$